

MATH 252Z (CRN: 13210), FALL 2025: CALCULUS II

Lectures. MTWF 9am-9:50m at 105 University

Instructor. Dr. Xiaojing Chen–Murphy

- Email: xchenm@uoregon.edu
- Office Hours: Mondays at 1-1:50pm, Tuesdays and Fridays at 12-12:50 in Tykeson 333

Course Material.

- *Textbook and Materials:* OpenSTAX Calculus Volume II. An electronic edition of this text is available for free at <http://openstax.org/details/books/calculus-volume-2>.
- *Calculators:* Calculators are not allowed in any of the quizzes or exams.
- *Prerequisite:* C– or better in Math 251.
- *Course Content:* This course begins with a brief review on antiderivatives in Section 4.10 of OpenSTAX Calculus Volume I (<https://openstax.org/details/books/calculus-volume-1>), and then covers most of Chapters 1–3 in the OpenSTAX Calculus Volume II. Topics covered in this course are listed as follows: antiderivatives and notion of integrals; Riemann sum computation; the Fundamental Theorem of Calculus, and various techniques for calculating integrals, including substitution, integration by parts, trigonometric integrals and trigonometric substitution; numerical integrals, and improper integrals; applications of the integral such as computing areas, and volumes, applications in physics, and application in probability.

Homework, Quizzes and Exams.

- **WebWork Homework (10%).** WebWork homework assignments are due every Tuesday and most Fridays, and will be closed two days after their due dates.
 - The period after the homework due date and before it is closing date is referred as “Reduced Scoring Period” on WebWork. Homework problems that are completed during reduced scoring period will result in a 20% penalty. One lowest WebWork homework will be dropped.
- **Quizzes (12%).** There will be weekly quizzes on Wednesdays.
 - Quizzes are about 10-minute long. Quiz may take place at any time during the 50-minute period of lecture. No make-up time will be given if you miss some or the whole quiz because of absence or arriving late or leaving early.
 - Quiz questions are based on the materials covered up to the Tuesday before that quiz. A list of topics of quiz questions will be shared on Canvas by the end of the Mondays.
 - One lowest quiz will be dropped. This is a uniform way of handling illness, family emergency and any other events that may interfere with the course. Thus requests to take quizzes at a time different from the scheduled quiz time will not be granted.
- **Attendance (3%).** This is an in-person class. If you have to miss a lecture for emergency, and you inform me your absence **before** the lecture, then the absence will be excused (i.e. not counted against your attendance record).
 - 3% credit - two or less absences
 - 2% credit - three or four absences
 - 1% credit - five or six absences
 - 0% credit - more than six absences

- **Midterm 1 (20%) - Friday, October 24**, 50-minute exam during class time.
- **Midterm 2 (20%) - Friday, November 21**, 50-minute exam during class time.
- **Final Exam (35%) - Tuesday, December 9 starting at 5pm**, 120-minute exam. The final is cumulative.

Quiz and Exam Rules.

- All quizzes and exams are closed-note, closed-book, and without electronic assistance (including calculators, cell phones, tablets, computers, and etc). Using any notes or electronic device or communicating with anyone except me during an exam constitutes cheating. Cheating on a quiz or exam will be reported to the university with a recommended punishment of failing the course.
- All exams are in person exams in the regular lecture classroom. The attendance of quizzes and exams is mandatory. Requests to take quizzes, midterm exams or the final exam at a time different from the scheduled quiz/exam time will not be granted. Missed midterms will be handled on a case by case basis.
- If you miss an exam, you will get zero for that exam. However, you may replace your lower midterm score by the final exam score if the final exam score is higher. Note that this is a uniform way of handling absence of one midterm due to illness, family emergency and any other events. Thus requests to take the midterm exams or the final exam at a time different from the scheduled exam time will not be granted.
- If you miss the final exam and are otherwise passing the class, you may request the incomplete grade for the class. Rules for eligibility of requesting incomplete grade is strict by the university policy. You may refer to this link <https://provost.uoregon.edu/grades-incompletes-policy> for the University rules regarding incomplete grades.

Grading Policy. On all written work (e.g. quizzes and exams) submitted for grading, you are expected to show your work clearly and completely; you will be graded on your work as well as on your final answers, but a correct final answer that is neither supported nor justified by work you have shown will receive no credit. I am happy to address any questions regarding the grading of any submitted work, and I encourage you to ask. However, you must bring such questions to my attention within one week of the date it is returned to the class or no changes will be made in the grade. Your final grade of the course is computed according to the following scale:

Homework (10%) + Attendance (3%) + Quiz (12%) + Midterm 1 (20%)

+ Midterm 2 (20%) + Final (35%)

Your final percentage will be rounded to the nearest whole number. The letter grade will be assigned based on the following table. Note that the instructor reserves the right to adjust grades at the end of the term based on the outcomes of the course.

Percentage	Grade	Percentage	Grade	Percentage	Grade	Percentage	Grade	Percentage	Grade
98%–100%	A+	88%, 89%	B+	78%, 79%	C+	68%, 69%	D+	0 – 59%	F
93%–97%	A	83%–87%	B	73%–77%	C	63%–67%	D		
90%– 92%	A–	80%–81%	B–	70%–72%	C–	60%–62%	D–		

Note: Any questions regarding grades will not be answered or discussed via email. You will need to meet with me to discuss any concerns or questions about your grades.

Classroom Expectations.

- Students are expected to behave respectfully towards each other and toward the instructor during class time. This includes arriving class on time, and not leaving the class early.
- You may use your tablets or computer to take notes. Otherwise, cell phones, and any electronic devices are not permitted for any activities that are unrelated to the course during lecture time.
- You should read the sections in the textbook and preview the posted slides before they are covered in the class.
- Arrive before class begins and exit only after class is dismissed. It is distracting to have students entering and exiting, so plan accordingly. If you have to miss a class, you are responsible to make up all the materials you miss.

Canvas. Canvas is the place to go for course information and engagement outside of class time. You are responsible for any announcements made on Canvas. To access our course Canvas site, log into <https://canvas.uoregon.edu> using your DuckID. If you have questions about using Canvas, visit the Canvas support page. Canvas and Technology Support also is available by phone (541-346-4357) or by live chat on the Live Help webpage.

Learning Environment. The University of Oregon and I are dedicated to fostering inclusive learning environments for all students and welcomes students with disabilities into all of the University's educational programs. The Accessible Education Center (AEC) assists students with disabilities in reducing campus-wide and classroom-related barriers. If you have or think you have a disability (<https://aec.uoregon.edu/content/what-disability>) and experience academic barriers, please contact the AEC to discuss appropriate accommodations or support. Visit 360 Oregon Hall or <https://aec.uoregon.edu> for more information. You can contact AEC at 541-346-1155 or via email at uoaec@uoregon.edu.

Academic Conduct. All students are expected to maintain academic integrity. Cheating will not be tolerated. Cheating includes but is not limited to use of any unauthorized assistance in taking quizzes, tests, or examinations or knowingly providing any unauthorized assistance to another student on quizzes, tests, or examinations. The University Student Conduct Code (<https://dos.uoregon.edu/code-procedures>) defines academic misconduct. Students are prohibited from committing or attempting to commit any act that constitutes academic misconduct. By way of example, students should not give or receive (or attempt to give or receive) unauthorized help on assignments or examinations without express permission from the instructor. Students should properly acknowledge and document all sources of information (e.g. quotations, paraphrases, ideas) and use only the sources and resources authorized by the instructor. If there is any question about whether an act constitutes academic misconduct, it is the students' obligation to clarify the question with the instructor before committing or attempting to commit the act. Additional information about a common form of academic misconduct, plagiarism, is available at <https://researchguides.uoregon.edu/citing-plagiarism>. All suspected cases of academic misconduct will be reported to the Dean of Students' Office of Student Conduct and Community Standards.

AI Policy. Students may use GenAI tools in this class to help with certain aspects of course work and assignments. This includes brainstorming ideas, creating a paper outline, or summarizing research findings of articles. However, you cannot use content such as text or graphics created by GenAI tools in your work; rather, you must be the author/creator of your work submissions. For example, you can use a GenAI tool to suggest a paper outline based on a draft you provide it, but you cannot submit a paper with text generated by GenAI as if the text is your own writing. As this is a math class, it is not appropriate to use GenAI to solve a math problem. If you did use GenAI for any of the approved aspects above, you must include a screenshot from what GenAI

provided you. Be advised, in accordance with UO policy, if I believe you've handed in work created whole or in part by GenAI tools, I may submit a report of suspected academic misconduct to the Office of Student Conduct and Community Standards for that office to make a determination of responsibility and, if warranted, assess a grade penalty. So, if you are in doubt or have questions about a particular GenAI tool and if its use is okay, check in with me and let's discuss!

Inclement Weather. It is generally expected that class will meet unless the University is officially closed for inclement weather. If it becomes necessary to cancel class, move the class online (via Zoom), or reschedule an exam while the University remains open, this will be announced on Canvas and by email. Updates on inclement weather and closure are also communicated as described on <https://hr.uoregon.edu/content/inclement-weather-immediate-updates>.

Course Learning Outcomes.

- 1). Approximate definite integrals using Riemann sums and apply this to the concept of accumulation and the definition of the definite integral.
- 2). Explain and use both parts of the Fundamental Theorem of Calculus.
- 3). Choose and apply integration techniques including substitution (including u -substitution and trigonometric substitution), integration by parts, basic partial fraction decomposition, and numerical techniques to integrate combinations of power, polynomial, rational, exponential, logarithmic, trigonometric, and inverse trigonometric functions.
- 4). Use the integral to model and solve problems in mathematics involving area, volume, net change, average value, and improper integration.
- 5). Apply integration techniques to solve a variety of problems, such as work, force, center of mass, or probability.